Date: Roll No.

III SEMESTER B.Tech (ECE, EVDT, EIOT)

MID-SEMESTER EXAMINATION SEPTEMBER, 2026

Course Code- ECECC302 /EIECC302/ VTECC302 Course Title- Signals & Systems

Time: 1:30 Hrs Max Marks: 20

Attempt all questions. Missing data / information (If any) may be suitably assumed & mentioned in the answer.

Q. No.		Questions	Marks	CO
Q 1	a	Express the following functions in terms of shifted/flipped versions of the unit step function [i.e $u(\pm t - t_0)$ *OR* $u[n-N]$], i. $x(t) = u(2t+6)$ ii. $y(t) = u\left(-\frac{t}{4}+2\right)$ iii. $z[n] = u[3n-7]$ In each case sketch the function.	03	CO1
	b	If $\int_{-\infty}^{\infty} y(\tau)\, x(t-\tau)\, d\tau = \sum_{k=0}^{n} x(t-kT_0)$, then y(t) is	01	CO1
Q 2	a	Consider a continuous-time system with input x(t) and output y(t) related by y(t) = 3x(sin t). Is the system stable, causal & time invariant	03	CO1
	b	Compute the time period of $x[n] = e^{\frac{j2\pi n}{10}} + e^{\frac{j6\pi n}{10}} + e^{\frac{j2\pi n}{15}} + e^{\frac{j2\pi n}{30}}$	01	CO1
Q 3	a	A continuous-time LTI system has an impulse response h(t) defined as: $h(t) = e^{at}u(t-1) + e^{bt}u(-t+1)$ where a and b are real-valued constants. i Determine the value(s) of a, b required for this system to be causal. ii Determine the exact range of values for a and b such that the system is BIBO stable iii Compute the step response of the system for a=0.5, b=2.	03	CO2
	b	A bounded periodic signal with time period T is applied as input to an LTI system with impulse response $h(t) = e^{-\lvert t\rvert}$. Is the output also periodic and bounded?	01	CO2
Q 4	a	An LTI system's impulse response h[n] is unknown except for two facts: a. it is causal and has finite duration of exactly 3 samples, b. when x[n] = δ[n] + δ[n−2] is applied, y[n] = {1, 0, 3, 0, 2} for n = 0 to 4. i Reconstruct h[n]. ii Compute the output when the input $z[n] = na^n u[n]$	03	
	b	Determine whether the discrete-time LTI system defined by the input-output relationship $y(n) = \sum_{k=n-2}^{n+2} x[k]$ is causal. Justify your answer based on the impulse response of the system.	01	CO1
Q 5	a	Consider a continuous-time LTI system characterized by the impulse response $h(t) = e^{-2t}u(t)$. An input signal $x(t) = e^{j6\pi t} \cdot \sum_{k=-\infty}^{\infty} \delta(t-k)$ is applied as input to the system i Is x(t) periodic? ii If Periodic, compute FS of input and output	03	CO1
	b	A continuous-time periodic signal x(t) has a Fourier series representation with Fourier coefficients X_k and a time period T_0=2. Determine the Fourier series representation of y(t)=x(2t-3).	01	CO1